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European option valuation with modified Heston model under stochastic market liquidity

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Abstract:

Comprehensive modeling of stock prices plays a critical role in financial derivatives valuation. Although, the classic Heston model captures the stochastic volatility, it struggles to replicate the term structure of implied volatility and assumes a perfectly liquid market. To address these inconsistencies, we introduce a Heston model by incorporating independent Ornstein–Uhlenbeck processes for both the long-term mean of volatility and stochastic market liquidity. Moreover, we derive an analytical European option pricing formula and validate the model using real market data from leading companies. We employ the root mean square error to assess the superiority of the introduced model against three benchmark models in replicating European option prices and implied volatility.

Keywords: European option pricing, Heston model, Stochastic liquidity, Stochastic volatility

Mathematics Subject Classification (2020): 91G20, 91G30, 60J76.

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1 Introduction

In 1993, Heston introduced his eponymous model overcorrected the limitations of the Black–Scholes–Merton (BSM) model in capturing the implied volatility patterns observed in equity mar-

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ket. The Heston model employs the Cox–Ingersoll–Ross stochastic differential equation (SDE) to cover the stochastic volatility. Although the Heston model remedied some limitations of the BSM model, it still produced inaccurate results when replicating European option prices and implied volatility. He and Chen (2021) extended the Heston model by including the assumption of a stochastic long-term mean for volatility. They incorporated an Ornstein–Uhlenbeck (OU) process to model the long-term mean of volatility, which in turn led to an improvement in the accuracy of the assessed implied volatility term structure. However, they presented their accuracy of the model from the unrealistic assumption of perfect market liquidity. Amihud and Mendelson (1986) highlighted the effect of liquidity risk on stock prices. Feng et al. (2014) represented the stochastic market liquidity framework utilizing an OU process to model market liquidity. Moreover, they employed this methodology for European options valuation in Feng et al. (2016). Following the same assumption, Cai et al. (2024) utilized Lévy processes to price vulnerable spread options. He et al. (2025) included credit risk in this framework for more accurate option valuation. Yahyavi and Modarresi (2026) investigated the accuracy of the Heston model for pricing European options, considering the existence of jumps in stock prices and the stochasticity of market liquidity.

In this paper, we present a modified version of the Heston model that incorporates both a stochastic long-term mean for volatility and the stochasticity of market liquidity. To capture these elements, we employ a Cox-Ingersoll-Ross (CIR) model for stochastic volatility, and two OU processes to model the stochastic long-term mean and stochastic market liquidity, respectively. Following the approach of Duffie et al. (2000), we derive the characteristic function of stock price returns. Then, we utilize the relationship between the characteristic function and the Cumulative Distribution Function (CDF) to achieve a closed-form formula for European option pricing.

Within the empirical analysis, we utilize real market European option price data from famous companies as Apple Inc., Amazon Inc., JPMorgan Chase & Co., and Walmart Inc. to evaluate the performance of proposed model against several benchmarks. As comparative models, we employ the classic Heston model which captures only stochastic volatility, the model which accounts only stochastic liquidity, and the model which incorporates both stochastic volatility and a stochastic long-term mean. We use the root mean square error (RMSE) to assess the accuracy of each model in replicating observed option prices and implied volatilities.

The remainder of this paper is organized as follows. Section 2 presents the SDEs of the modified Heston model under the risk-neutral measure and the corresponding characteristic function. Section 3 includes the derivation of an analytical formula for European option pricing. In Section 4, we report the results of the comparison between the proposed model and the benchmark models. Finally, the last section summarizes our findings and concludes the paper.

2 Modified Heston model

Let $\left(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{Q}\right)$ denote the filtered complete probability space, where $\mathbb{Q}$ is the risk-neutral measure and it is endowed with the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ where $t \in [0, T]$ for a fixed expiration time T. We define the model dynamics with respect to measure $\mathbb{Q}$ as below.

$$\begin{cases} \frac{dS_t}{S_t} = rdt + \sqrt{\nu_t}dB_{1,t} + \beta a_t dB_{2,t}, \\ d\nu_t = \kappa(\theta_t - \nu_t)dt + \sigma_1\sqrt{\nu_t}dB_{3,t}, \\ d\theta_t = \alpha dt + \sigma_2 dB_{4,t}, \\ da_t = \kappa_a(\theta_a - a_t)dt + \sigma_3 dB_{5,t}, \end{cases} \tag{2.1}$$

where r is the risk-free rate of return, ν_t is the stochastic volatility, κ is mean reversion speed, θ_t is the stochastic mean level of volatility, σ_1 is volatility of volatility, α is the drift of mean level of volatility, σ_2 is the volatility of mean level of volatility, a_t is stochastic market liquidity, κ^a is mean reversion speed of market liquidity, θ^a is mean level of market liquidity and σ_3 is volatility of market liquidity. All Brownian motion processes are mutually independent under measure $\mathbb{Q}$ except $B_{1,t}$ and $B_{3,t}$ with correlation ρ and $B_{2,t}$ and $B_{5,t}$ with correlation ρ_a.

Remark 2.1. *If $\beta = \alpha = \sigma_2 = \kappa_a = \theta_a = \sigma_3 = 0$, the proposed model reduces to the Heston model. If $\alpha = \sigma_2 = \kappa = \sigma_1 = 0$, the proposed model reduces to the Feng et al. (2014) model. And, if $\beta = \kappa_a = \theta_a = \sigma_3 = 0 = 0$, the proposed model reduces to the He and Chen (2021) model.*

By assuming $X_t := \ln(S_t)$, we derive the characteristic function of X_t employing the methodology presented by Duffie et al. (2000) for affine jump-diffusion models. The exponential-affine representation of the characteristic function is presented in the following theorem.

Theorem 2.2. *Let the underlying stock price follow the model specified in (2.1), then, the characteristic function of $X_t = \ln(S_t)$ is given by*

$$\phi(u, t, T; x, \nu, a, \theta) = \exp\left\{A(u;\tau) + B(u;\tau)a + C(u;\tau)a^2 + D(u;\tau)\nu + E(u;\tau)\theta + iux\right\}, \tag{2.2}$$

where $i = \sqrt{-1}$, $\tau = T - t$, T is the maturity time of the European option, and the coefficient *functions are*

$$
\begin{aligned}
A &= \frac{1}{4}\left(\left(a_1 - 2(i\sigma_3\beta\rho_a u - \kappa_a)\right)\tau - 2\ln\left(\frac{1-b_1e^{a_1\tau}}{1-b_1}\right)\right) \\
&\quad + \int_0^\tau \left(\frac{1}{2}\sigma_3^2B^2(u;s) + \kappa_a\theta_aB(u;s) + \frac{1}{2}\sigma_2^2E^2(u;s) + \alpha E(u;s)\right)ds + riu\tau, \\
B &= -\frac{\kappa_a\theta_a\left(a_1 - 2(i\sigma_3\beta\rho_a u - \kappa_a)\right)}{a_1\sigma_3^2}\frac{\left(1-e^{\frac{1}{2}a_1\tau}\right)^2}{1-b_1e^{a_1\tau}}, \\
C &= \frac{a_1 - 2(i\sigma_3\beta\rho_a u - \kappa_a)}{4\sigma_3^2}\frac{1-e^{a_1\tau}}{1-b_1e^{a_1\tau}}, \\
D &= \frac{a_2 - (i\sigma_1\rho u - \kappa)}{\sigma_1^2}\frac{1-e^{a_2\tau}}{1-b_2e^{a_2\tau}}, \\
E &= \frac{\kappa}{\sigma_1^2}\left[\left(a_2 - (i\sigma_1\rho u - \kappa)\right)\tau + 2ln\left(\frac{1-b_2e^{a_2\tau}}{1-b_2}\right)\right],
\end{aligned}
$$

and

$$
\begin{aligned}
a_1 &= 2\sqrt{(i\sigma_3\beta\rho_a u - \kappa_a)^2 + \sigma_3^2\beta^2(iu+u^2)}, \\
a_2 &= \sqrt{(i\sigma_1\rho u - \kappa)^2 + \sigma_1^2(iu+u^2)}, \\
b_1 &= \frac{2(i\sigma_3\beta\rho_a u - \kappa_a) - a_1}{2(i\sigma_3\beta\rho_a u - \kappa_a) + a_1}, \\
b_2 &= \frac{(i\sigma_1\rho u - \kappa) - a_2}{(i\sigma_1\rho u - \kappa) + a_2},
\end{aligned}
$$

where all parameters are defined in the model represented in (2.1). In the following section, we employ the characteristic function represented in Theorem 2.2 to derive an analytic formula for European option valuation.

3 European option price formula

To derive the European option price, we utilize the relation between the characteristic function and the CDF in the following theorem.

Theorem 3.1. *Given an arbitrary random variable X with Lebesgue integrable density function f and characteristic function ϕ, if the mean of X exists then the following relation holds between characteristic function ϕ and F as the CDF of X*

$$
F(x) = \frac{1}{2} - \frac{1}{2\pi}\int_0^\infty \Delta\left[\frac{\phi(u)\,e^{-iux}}{iu}\right]du, \tag{3.1}
$$

where $\Delta g(u) = g(u) + g(-u)$.

By following the risk-neutral pricing methodology, the price of an European call option with strike price K, maturity T, and underlying stock S_t is formulated as follows.

$$
C(S_t, K, \tau, \nu_t, a_t) = e^{-r\tau}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+|\mathcal{F}_t^{agg}\right], \tag{3.2}
$$

where $\tau = T - t$ and $\mathcal{F}_t^{agg} = \left\{\mathcal{F}_t^{B_{1,t}} \vee \mathcal{F}_t^{B_{2,t}} \vee \mathcal{F}_t^{B_{3,t}} \vee \mathcal{F}_t^{B_{4,t}} \vee \mathcal{F}_t^{B_{5,t}}\right\}_{t\geq 0}$. We are able to redefine the European option price by employing the relation (3.1) in Equation (3.2) as below.

$$C(I_1, I_2, \tau) = e^{-r\tau}\left[I_1 - KI_2\right], \tag{3.3}$$

where

$$I_1 = \phi(-i, t, T; x, \nu, a, \theta)\left[\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left(\frac{e^{-iu\ln(K)}}{iu}\frac{\phi(u-i, t, T; x, \nu, a, \theta)}{\phi(-i, t, T; x, \nu, a, \theta)}\right)du\right],$$
$$I_2 = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left(\frac{e^{-iu\ln(K)}}{iu}\phi(u, t, T; x, \nu, a, \theta)\right)du,$$

and $\phi(u, t, T; x, \nu, \alpha, \theta)$ is the characteristic function derived in Theorem 2.2. In the following section, we employ the analytic formula in (3.3) to compare the accuracy of the proposed model in replicating real market European option prices and implied volatility against benchmarks.

4 Model's accuracy comparison

We utilize real market data from European option chains for four distinct stocks, Apple Inc., Amazon Inc., JPMorgan Chase & Co., and Walmart Inc. We also employ the RMSE function to compare the efficiency of the represented model against benchmarks. The features captured by each benchmark model are briefly discussed in Table 1. As illustrated in Figures 1 and 2, the proposed model exhibits the lowest RMSE values when compared with benchmark models in replicating European option prices and implied volatility, respectively. The proposed model demonstrates enhanced accuracy in replicating the prices of European option chains, even when the chain is dense with multiple strike prices, as shown in Panel a) of Figure 1. Furthermore, as observed in Figure 2, the introduced model effectively captures the volatility smile curve, particularly in the at-the-money region.

Table 1: Benchmark models discussion.

Model	Stochastic volatility	Stochastic liquidity	Stochastic long-term mean
Heston	✓	×	×
Feng	×	✓	×
HeChan	✓	×	✓
Ours	✓	✓	✓

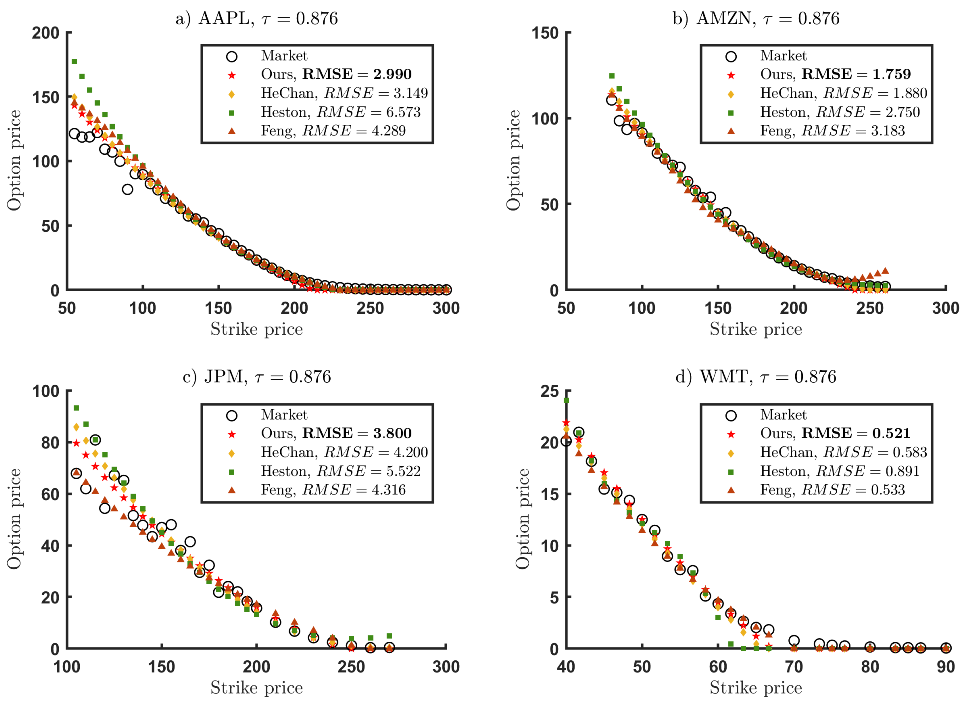

Figure 1: Comparison of the proposed model's replication of European option prices against benchmarks.

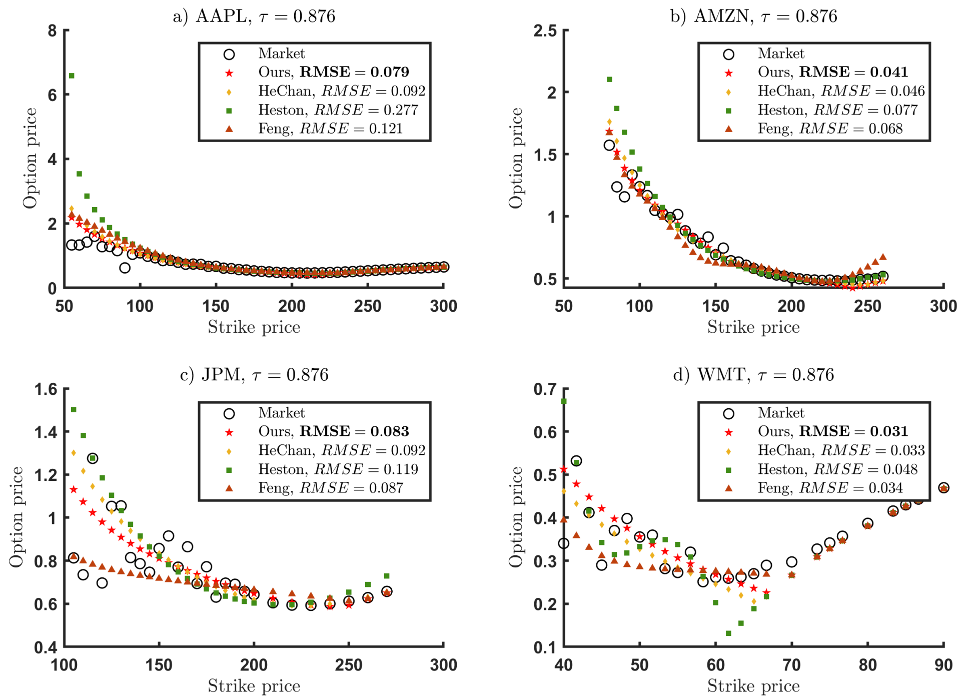

Figure 2: Comparison of the proposed model's replication of implied volatility against benchmarks.

Discussion and Conclusion

In this paper we have introduced a comprehensive model for pricing European options, integrating stochastic volatility, long-term mean of volatility, and market liquidity simultaneously. Furthermore, we have utilized real market data to perform a comparison between the introduced model and benchmarks. The empirical analysis results demonstrate the superiority of proposed model in achieving the lowest RMSE values.

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